

REAL-TIME WAVEFRONT RECONSTRUCTION PIPELINE

for Adaptive
Optics
Systems

Team Name: Motta Puffs

Problem Statement: Developing a high-performance, 4-stage pipeline for atmospheric turbulence characterization and real-time correction in Adaptive Optics systems.

TEAM MEMBERS

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TURBULENCE CHALLENGES

Kolmogorov Statistics

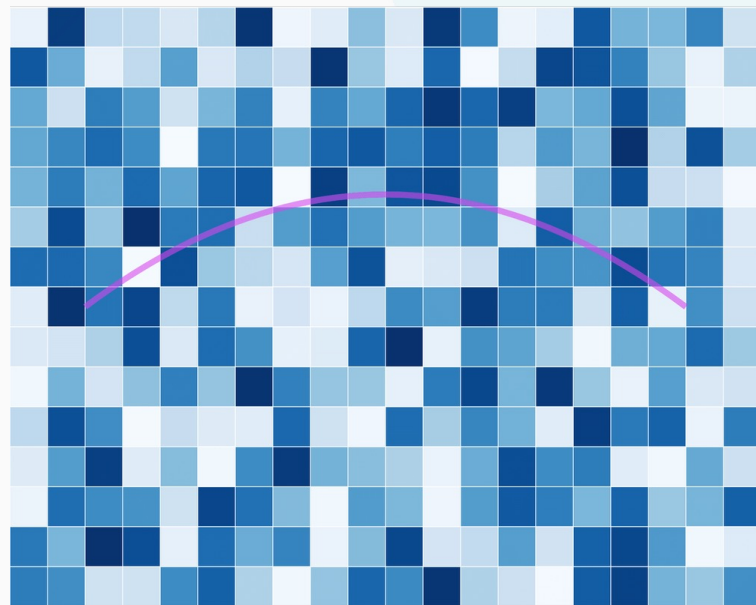
Phase aberrations follow $D_{\phi}(r) = 6.88 (r/r_0)^{5/3}$, necessitating precise statistical modeling of the atmosphere.

Taylor's Frozen Flow

Turbulence advects with wind, creating a dynamic environment that requires millisecond-level processing.

Phase Degradation

Uncorrected turbulence reduces the Strehl ratio, impacting image resolution and data throughput significantly.



4-STAGE PIPELINE ARCHITECTURE

01

INGESTION & CENTROIDING

High-speed BMP loading. **C + OpenMP** optimized hot-path for sub-millisecond centroiding.

02

WAVEFRONT RECONSTRUCTION

Modal (Zernike) and **Zonal (Fried)** reconstruction from slope vectors using interaction matrices.

03

TURBULENCE ANALYSIS

Statistical estimation of **r_0** , **τ_0** , and wind velocity for real-time system tuning.

04

CONTROL & CORRECTION

Mapping phase to **DM Actuators**. Closed-loop simulation achieving >0.99 Strehl ratio.

STAGE 1: FAST CENTROIDING

ALGORITHMIC VERSATILITY

Supports Center of Gravity (CoG), Thresholded-CoG (T-CoG), and Windowed-CoG (W-CoG) to handle varying noise regimes and background pedestals.

C-ENGINE OPTIMIZATION

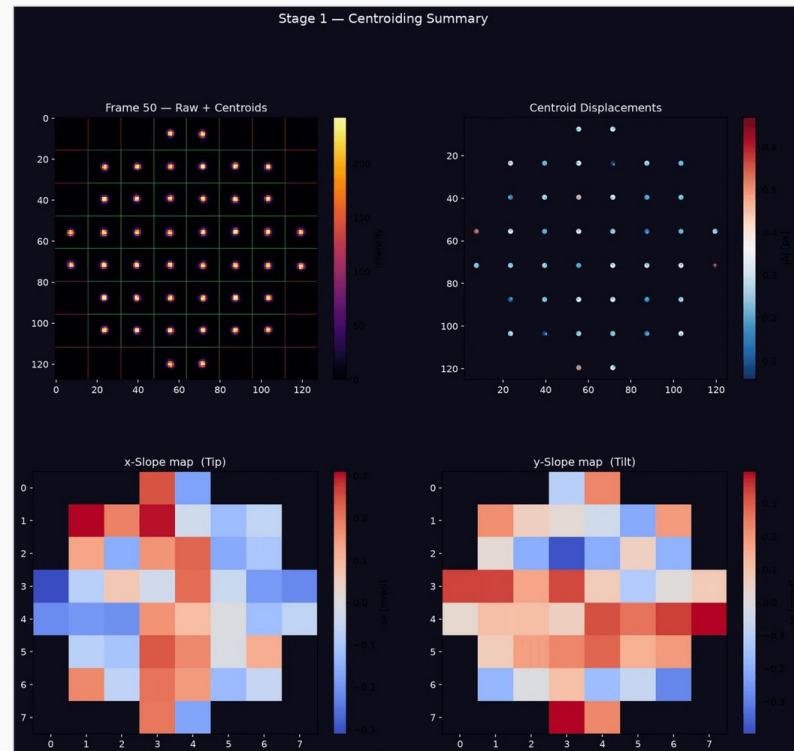
Hot-path operations implemented in C with OpenMP multi-threading, achieving sub-millisecond processing times per frame.

PUPIL MASKING

Automated sub-aperture grid generation with support for circular and annular telescope apertures, filtering invalid lenslets.

DIAGNOSTIC MONITORING

Real-time visualization of centroid displacements and angular slope maps (Tip/Tilt) for immediate pipeline validation.



WAVEFRONT RECONSTRUCTION

MODAL DECOMPOSITION

Reconstructing wavefronts using Zernike polynomials to identify specific optical aberrations like Tip, Tilt, and Defocus.

FRIED GEOMETRY

Implementing least-squares zonal reconstruction for high-order spatial correction across the sub-aperture grid.

INTERACTION MATRIX

Pre-computed reconstructor matrices enable rapid conversion from slope vectors to phase maps in real-time.

PERFORMANCE METRICS

Continuous tracking of Strehl ratio and slope residual RMS to validate reconstruction accuracy and loop stability.

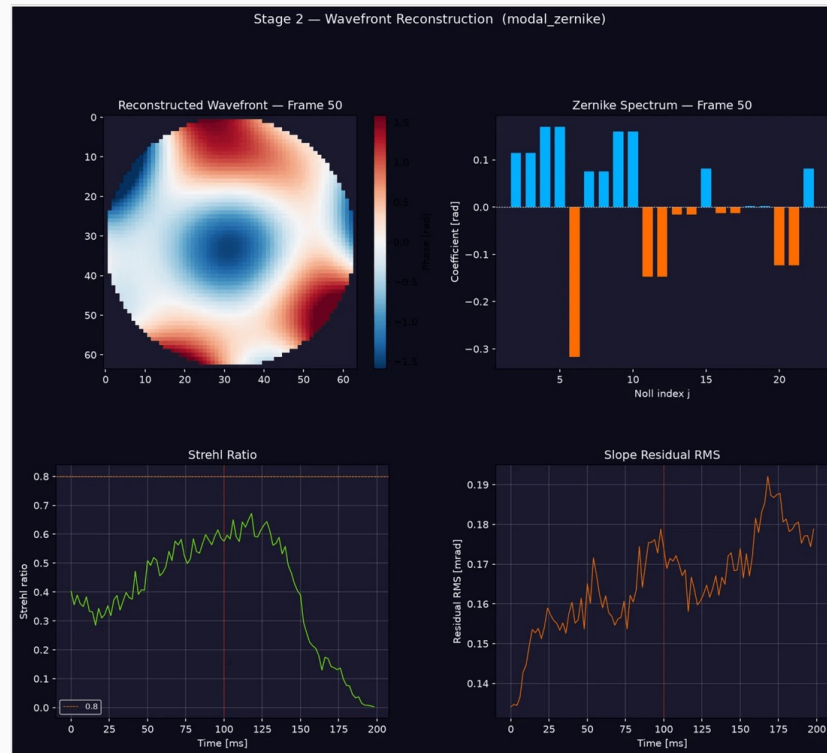


Figure: Reconstructed Wavefront, Zernike Spectrum, and Strehl Ratio Performance

TURBULENCE CHARACTERIZATION

FRIED PARAMETER (r_0)

Real-time estimation of atmospheric coherence length from Zernike variance and spatial structure functions.

TEMPORAL ANALYSIS

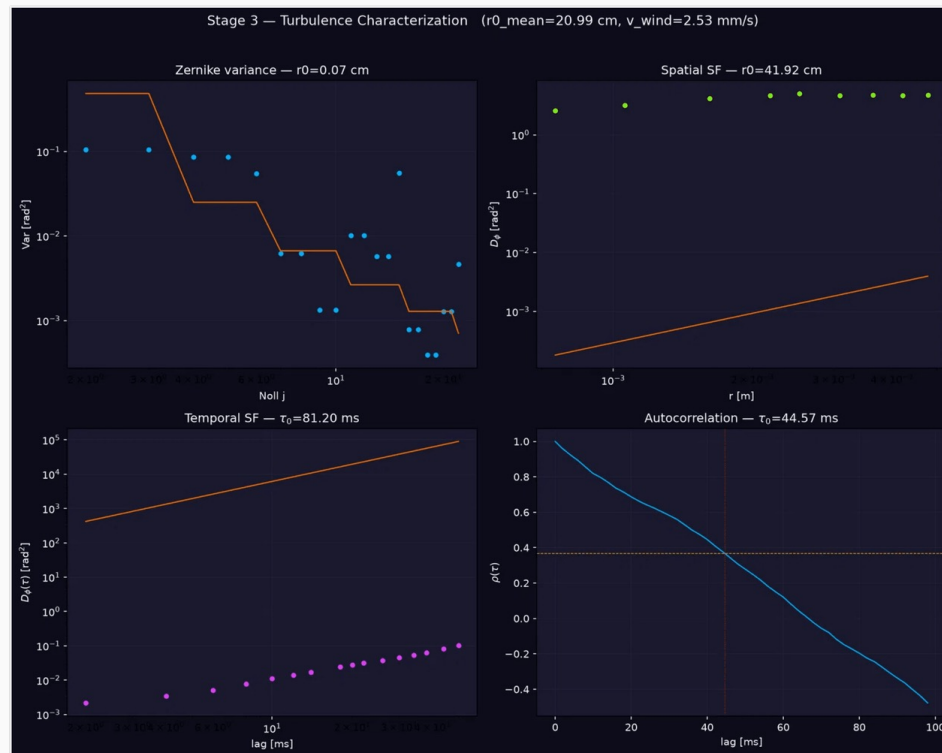
Calculation of coherence time (τ_0) and wind velocity using temporal autocorrelation of phase screens.

MODAL POWER SPECTRUM

Analyzing energy distribution across Zernike modes to validate Kolmogorov statistics compliance.

SYSTEM OPTIMIZATION

Providing critical feedback for AO loop gain tuning and exposure time adjustments.



DM CONTROL & CORRECTION

ACTUATOR MAPPING

Precise mapping of reconstructed phase maps to Deformable Mirror (DM) actuator stroke commands using the Fried geometry.

INFLUENCE FUNCTIONS

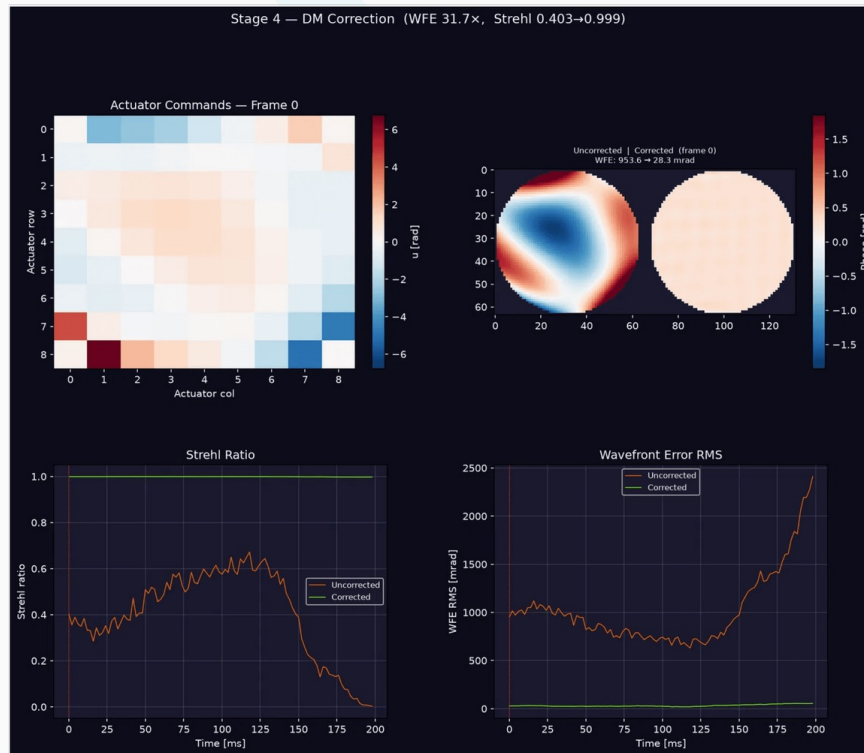
Modeling the response of individual actuators to ensure smooth and accurate wavefront compensation across the pupil.

CLOSED-LOOP READINESS

Architecture designed for seamless integration with real-time hardware controllers and high-speed DM interfaces.

Strehl Ratio: 0.40 → 0.99+

Demonstrated near-perfect correction in simulation.



TECHNICAL STACK

CORE LANGUAGES

Python 3.12

High-level orchestration, statistical analysis, and visualization using NumPy and SciPy.

C (ISO C99)

Performance-critical centroiding engine and matrix operations for real-time compliance.

OPTIMIZATION

OpenMP

Multi-threaded parallel processing for sub-aperture centroiding calculations.

SIMD Vectorization

Compiler-optimized math loops for high-throughput pixel processing.

STANDARDS

YAML / NPZ

Flexible configuration management and efficient binary data interchange formats.

BMP Frames

Standardized raw detector frame ingestion supporting 8-bit and 16-bit grayscale.

Validation Strategy: *End-to-end pipeline verification using a custom **Synthetic Turbulence Generator** implementing Kolmogorov phase screens and Taylor frozen flow.*

SCALABILITY & IMPLEMENTATION



COMPUTATIONAL COST

Achieves **sub-millisecond latency** per stage on standard workstations. Highly parallel architecture is ready for **FPGA or GPU** acceleration for large-scale MLA grids.



HARDWARE AGNOSTIC

Modular design supports diverse **SH-WFS sensors** and **DM architectures**. All parameters are managed via flexible YAML-based configuration files.



FUTURE ROADMAP

Planned integration of **Predictive Control** algorithms to mitigate temporal lag and support for **Multi-Conjugate AO** (MCAO) systems.

THANK YOU

WAVEFRONT RECONSTRUCTION PIPELINE

Repository: github.com/Croionxx/Wavefront-construction

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